

Stanisław Apanasiewicz 198169 – Task 3

Galaxies, Stars, Planets etc

Query1

Query1.sql ...198169 (87)

```
1  -- group the smaller celestials (moons and planets) into 4 subgroups
2  -- gas planets, solid planets, moons with shorter orbit periods (shorter than our moon) , moons with longer orbit periods
3  CREATE OR ALTER VIEW Celestial_Types AS
4  SELECT body_name, mass_scientific_notation, mass_exp_10, diameter, surface_temperature,
5         CASE
6         WHEN planet_name IS NOT NULL
7         AND state_of_matter LIKE 'Gas'
8         THEN 'Gas Planet'
9         WHEN planet_name IS NOT NULL
10        AND state_of_matter LIKE 'Solid'
11        THEN 'Solid Planet'
12        WHEN moon_name IS NOT NULL
13        AND moon_orbit_time >= 1
14        THEN 'Long day Moon'
15        WHEN moon_name IS NOT NULL
16        AND moon_orbit_time < 1
17        THEN 'Short day Moon'
18        ELSE 'error'
19        END AS celestial_type
20 FROM Celestial_Body
21 LEFT JOIN Planet
22 ON Celestial_Body.body_name = Planet.planet_name
23 LEFT JOIN Moon
24 ON Celestial_Body.body_name = Moon.moon_name
25 WHERE planet_name IS NOT NULL
26 OR moon_name IS NOT NULL;
27
28 GO
29
30 SELECT celestial_type, COUNT(*) AS celestials_per_type
31 FROM Celestial_Types
32 GROUP BY celestial_type
33 ORDER BY celestials_per_type DESC
34 /* show how many of these types are in a database and
```

75 % Nie znaleziono żadnych problemów W: 13, Zn: 27, Kol: 33 MIESZANE CRLF Windows 1250

Wyniki Komunikaty

	celestial_type	celestials_per_type
1	Solid Planet	14
2	Long day Moon	12
3	Gas Planet	9
4	Short day Moon	2

Zapytanie zostało wykonane pomyślnie. sql198169server.database.wi... s198169 (87) s198169-space 00:00:00 Wiersz: 1, kolumna: 1 Wiersze: 4

Query2

```
1  CREATE OR ALTER VIEW MassRank AS
2  SELECT *,
3         ROW_NUMBER() OVER
4         (
5         PARTITION BY celestial_type
6         ORDER BY mass_exp_10 DESC, mass_scientific_notation DESC
7         ) AS bigRank,
8         ROW_NUMBER() OVER
9         (
10        PARTITION BY celestial_type
11        ORDER BY mass_exp_10 ASC, mass_scientific_notation ASC
12        ) AS smallRank
13 FROM Celestial_Types;
14 GO
15 /* Mass is in exponential notation so the view MassRank is used to order the data from the heaviest
16 (first to rank the exponent, then mantissa)
17 in a particular celestial type (1 is the heaviest)*/
18 SELECT
19     celestial_type,
20     ROUND(AVG(diameter), 0) AS average_diameter,
21     MAX(diameter) AS max_diameter,
22     ROUND(STDEV(diameter), 0) AS diameter_std,
23     ROUND(AVG(surface_temperature), 0) AS average_temperature,
24     MAX(surface_temperature) AS temperature_max,
25     ROUND(STDEV(surface_temperature), 0) AS temperature_std,
26     MAX(CASE WHEN bigRank = 1 THEN mass_scientific_notation END)
27     || ' * 10^' ||
28     CAST(MAX(mass_exp_10) AS VARCHAR(2))
29     || ' kg ' AS maximum_mass,
30     MAX(CASE WHEN smallRank = 1 THEN mass_scientific_notation END)
31     || ' * 10^' ||
32     MIN(mass_exp_10)
33     || ' kg ' AS minimum_mass
34 FROM MassRank
35 GROUP BY celestial_type;
36 /* group the smaller celestials (moons and planets) into 4 subgroups to see if there is any difference in their statistics
37 (mean, max and std for diameter and temperature), max and min for the masses
38 the query uses a view from the previous query( gas planets, solid planets, moons with shorter periods , moons with longer periods)*/
39 --SELECT * FROM MassRank
```

75 % 2 0 W: 36, Zn: 15 SPACJE CRLF Windows 1250

Wyniki Komunikaty

	celestial_type	average_diameter	max_diameter	diameter_std	average_temperature	temperature_max	temperature_std	maximum_mass	minimum_mass
1	Gas Planet	107249	250000	79395	538	2500	868	1.9 * 10 ²⁷ kg	2 * 10 ²⁵ kg
2	Long day Moon	2688	5268	1869	109	233	62	1.48 * 10 ²³ kg	1.47 * 10 ¹⁵ kg
3	Short day Moon	209	396	264	148	233	120	3.7 * 10 ¹⁹ kg	1.06 * 10 ¹⁶ kg
4	Solid Planet	12414	25000	6678	426	2000	474	8.63 * 10 ²⁴ kg	3.3 * 10 ²³ kg

Zapytanie zostało wykonane pomyślnie. sql198169server.database.wi... s198169 (85) s198169-space 00:00:00 Wiersz: 1, kolumna: 1 Wiersze: 4

Query3

```
1 SELECT Celestial_Body.body_name, date_of_discovery, diameter, Observer.observer_id, observer_name, observer_surname
2 FROM Observer
3 INNER JOIN Discovered
4 ON Observer.observer_id = Discovered.observer_id
5 INNER JOIN Discovery
6 ON Discovered.discovered_body_name = Discovery.discovered_body_name
7 INNER JOIN Celestial_Body
8 ON Discovery.discovered_body_name = Celestial_Body.body_name
9 LEFT JOIN Planet
10 ON Celestial_body.body_name = Planet.planet_name
11 LEFT JOIN Star
12 ON Celestial_body.body_name = Star.star_name
13 LEFT JOIN Moon
14 ON Celestial_body.body_name = Moon.moon_name
15 ORDER BY diameter DESC
16 -- show the names of observers who discovered the largest objects
```

75 % Nie znaleziono żadnych problemów W.: 9, Zn.: 17 TABULATORY CRLF Windows 1250

Wyniki Komunikaty

	body_name	date_of_discovery	diameter	observer_id	observer_name	observer_surname
1	M87	2019-04-10	38000000000	M1211828890	Janusz	Kosmiczny
2	Alpha Centauri A	2019-03-24	1700000	M3434424890	Sebastian	Alvarez
3	Alpha Centauri A	2019-03-24	1700000	M6767676767	Igor	Owczynski
4	Alpha Centauri B	2020-02-23	1200000	M3434424890	Sebastian	Alvarez
5	Alpha Centauri B	2020-02-23	1200000	M6767676767	Igor	Owczynski
6	WASP-12b	2018-04-01	250000	M3434424890	Sebastian	Alvarez
7	WASP-12b	2018-04-01	250000	M4434563331	John	Czarnodziury
8	Kepler-22b	2011-12-05	29000	M9299222399	Timothy	Cheese
9	Proxima Centauri c	2016-08-24	20000	M3434424890	Sebastian	Alvarez
10	Giese 581g	2010-09-29	19000	F1234567890	Agata	Gwiezdna
11	Kepler-452b	2015-07-23	18000	M1234324521	Jakub	Kaszdan
12	Kepler-186f	2014-04-17	13000	M1232828890	Michael	Satellarmy
13	Proxima Centauri b	2016-08-24	7000	M3434424890	Sebastian	Alvarez
14	TRAPPIST-1e	2017-02-22	5800	F6234522823	Paulina	Konstelama
15	Proxima Centauri d	2016-08-24	5000	M3434424890	Sebastian	Alvarez
16	GW150914	2015-09-14	370	M1234567822	Ryszard	Ziemsly
17	Cygnus X-1	1964-01-01	126	M6767676767	Igor	Owczynski

Zapytanie zostało wykonane pomyślnie. sql198169server.database.wi... s198169 (83) s198169-space 00:00:00 Wiersz: 1, kolumna: 1 Wiersze: 17

Query4

```
1 CREATE OR ALTER VIEW Densities AS
2 SELECT
3     body_name,
4     ROUND((mass_exp_10 + LOG10(mass_scientific_notation)), 2) AS log_of_mass,
5     diameter AS diameter_km,
6     ROUND (LOG10(diameter), 4) AS log10_of_diameter,
7     ROUND ( (mass_exp_10 + LOG10(mass_scientific_notation)) / ((LOG10((diameter)) + 3), 2) AS density_coefficient,
8     CASE
9         WHEN planet_name IS NOT NULL
10            THEN 'Planet'
11         WHEN star_name IS NOT NULL
12            THEN 'Star'
13         WHEN moon_name IS NOT NULL
14            THEN 'Moon'
15         WHEN black_hole_name IS NOT NULL
16            THEN 'Black_hole'
17         ELSE 'error'
18     END AS celestial_type,
19     CASE
20         WHEN planet_name IS NOT NULL
21            THEN state_of_matter
22         WHEN star_name IS NOT NULL
23            THEN star_type
24         WHEN black_hole_name IS NOT NULL
25            THEN B.type_
26         ELSE '-'
27     END AS Subtype
28 FROM Celestial_Body CB
29 LEFT JOIN Planet P
30 ON CB.body_name = P.planet_name
31 LEFT JOIN Moon
32 ON CB.body_name = Moon.moon_name
33 LEFT JOIN Black_Hole B
34 ON CB.body_name = B.black_hole_name
35 LEFT JOIN Star S
36 ON CB.body_name = S.star_name
37 GO
```

62 % Nie znaleziono żadnych problemów W.: 46, Zn.: 3 MIESZANE CRLF Windows 1250

Wyniki Komunikaty

	body_name	log_of_mass	diameter_km	log10_of_diameter	density_coefficient	celestial_type	Subtype
1	GRO J1655-40	31,1	38	1,5798	26,36	Black_hole	stellar
2	LMC X-3	31,15	42	1,6232	26,28	Black_hole	stellar
3	XTE J1118+480	31,15	42	1,6232	26,28	Black_hole	stellar
4	V404 Cygni	31,26	54	1,7324	26,06	Black_hole	stellar
5	LMC X-1	31,34	65	1,8129	25,9	Black_hole	stellar

	body_name	log_of_mass	diameter_km	log10_of_diameter	density_coefficient	celestial_type	Subtype
1	WOH G64	31,4	2140000000	9,3304	3,41	Star	supergiant
2	Betelgeuse	31,34	1234000000	9,0913	4,07	Star	supergiant
3	Antares	31,38	946000000	8,9759	4,45	Star	supergiant
4	M31-RV	30,9	500000000	8,699	4,81	Star	supergiant
5	HV 2112	31,7	700000000	8,8451	5,16	Star	supergiant

Zapytanie zostało wykonane pomyślnie. sql198169server.database.wi... s198169 (75) s198169-space 00:00:00 Wiersz: 1, kolumna: 1 Wiersze: 10

```

39      GO
40      /* a query to show which exact types of celestial bodies tend to be the most dense, here are 5 most dense and 5 least dense
41      the density coefficient is not the exact density value, but its logarithm, it is done like this because database does not store large numbers such as 10^41,
42      its exponent and mantissa*/
43      SELECT TOP(5) * FROM Densities
44      ORDER BY density_coefficient DESC
45
46      SELECT TOP(5) * FROM Densities
47      ORDER BY density_coefficient ASC
48
49

```

83 % Nie znaleziono żadnych problemów W.: 46, Zn.: 3 MIESZANE CRLF Windows 1250

Wyniki Komunikaty

	body_name	log_of_mass	diameter_km	log10_of_diameter	density_coefficient	celestial_type	Subtype
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	body_name	log_of_mass	diameter_km	log10_of_diameter	density_coefficient	celestial_type	Subtype
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✓ Zapytanie zostało wykonane pomyślnie. sql198169server.database.wi... s198169 (75) s198169-space 00:00:00 Wiersz: 1, kolumna: 1 Wiersze: 10

Queryy5

```

1      CREATE OR ALTER VIEW Stars_and_Planets AS
2      SELECT *,
3      CASE
4          WHEN planet_name IS NOT NULL
5              THEN 'Planet'
6          WHEN star_name IS NOT NULL
7              THEN 'Star'
8      END AS celestial_type
9      FROM Celestial_Body CB
10     LEFT JOIN Planet P
11         ON CB.body_name = P.planet_name
12     LEFT JOIN Star S
13         ON CB.body_name = S.star_name
14     WHERE planet_name IS NOT NULL
15        OR star_name IS NOT NULL;
16     -- creating a view of only the planets and stars and their info from a superclass Celestial_Body
17     GO
18
19     SELECT body_name, diameter, mass_scientific_notation, ' * 10^' AS TEXT, mass_exp_10
20     FROM Stars_and_Planets AS SP
21     WHERE SP.celestial_type = 'Planet' AND diameter <=
22     (SELECT MIN(Stars_and_Planets.diameter)
23     FROM Stars_and_Planets
24     WHERE Stars_and_Planets.celestial_type = 'Star')
25     ORDER BY diameter DESC
26
27     -- selecting all planets which are not larger (in diameter) than the smallest star

```

91 % Nie znaleziono żadnych problemów W.: 22, Zn.: 40 MIESZANE CRLF Windows 1250

Wyniki Komunikaty

	body_name	diameter	mass_scientific_notation	TEXT	mass_exp_10
1	Proxima Centauri d	5000	1,55	* 10^	24
2	Mercury	4879	3,3	* 10^	23

✓ Zapytanie zostało wykonane pomyślnie. sql198169server.database.wi... s198169 (96) s198169-space 00:00:00 Wiersz: 1, kolumna: 1 Wiersze: 2

Query6

```

1
2 SELECT planet_name, semi_major_axis, state_of_matter, Planet.solar_system_name FROM Planet
3     INNER JOIN Celestial_Body
4     ON Planet.planet_name = Celestial_Body.body_name
5     INNER JOIN Solar_System
6     ON Planet.solar_system_name = Solar_System.solar_system_name
7 WHERE planet_name IN
8     (
9         SELECT planet_name
10        FROM Planet P
11        INNER JOIN Moon M
12        ON P.planet_name = M.parent_planet_name
13        GROUP BY planet_name
14        HAVING (COUNT(*) > 1)
15     )
16 AND semi_major_axis >
17     (
18         SELECT AVG(semi_major_axis)
19        FROM Planet
20        WHERE solar_system_name = 'The Solar System'
21     );
22
23 -- select all the planets which have multiple moons and have a semi major axis larger than the on average a planet in our solar system
24

```

91 % Nie znaleziono żadnych problemów W.: 1, Zn.: 1 TABULATORY CRLF Windows 1250

Wyniki Komunikaty

	planet_name	semi_major_axis	state_of_matter	solar_system_name
1	Saturn	9,582	gas	The Solar System
2	Uranus	19,218	gas	The Solar System

Zapytanie zostało wykonane pomyślnie. sql198169server.database.wi... s198169 (76) s198169-space 00:00:00 Wiersz: 1, kolumna: 1 Wiersze: 2

Query7

```

1 SELECT celestial_name,
2     COUNT(Monitor_Schedule.celestial_name) AS observatories_monitoring,
3     AVG(diameter) AS body_diameter
4 FROM Monitor_Schedule
5     INNER JOIN Celestial_Body
6     ON Monitor_Schedule.celestial_name = Celestial_Body.body_name
7 GROUP BY celestial_name
8 ORDER BY observatories_monitoring DESC, body_diameter ASC
9 -- select the Celestial bodies which are observed and sort them on which is observed the most

```

91 % Nie znaleziono żadnych problemów W.: 1, Zn.: 1 TABULATORY CRLF Windows 1250

Wyniki Komunikaty

	celestial_name	observatories_monitoring	body_diameter
1	Miranda	3	471
2	Phobos	2	22
3	Cygnus X-1	1	126
4	Triton	1	2706
5	Europa	1	3121
6	Mars	1	6779
7	Proxima Centauri b	1	7000
8	Kepler-186f	1	13000
9	Saturn	1	116460
10	WASP-12b	1	2500000
11	Sagittarius A*	1	240000000
12	Betelgeuse	1	1234000000

Zapytanie zostało wykonane pomyślnie. sql198169server.database.wi... s198169 (95) s198169-space 00:00:00 Wiersz: 1, kolumna: 1 Wiersze: 12

Query8

```

1  -- Selecting the planets with added info about the stars that they are surrounding
2  CREATE OR ALTER VIEW Stars_and_planets2 AS
3  SELECT star_name,
4         COUNT(*) OVER (PARTITION BY star_name) AS star_count -- counting how many times a star appears in this table
5         ,planet_name, orbit_time, semi_major_axis
6  FROM Planet P
7  INNER JOIN Solar_System SL
8  ON P.solar_system_name = SL.solar_system_name
9  INNER JOIN Star S
10 ON SL.center_star = S.star_name;
11 GO
12
13 SELECT *,
14        POWER(orbit_time, 2) AS 'T^2',
15        POWER(semi_major_axis, 3) AS 'R^3',
16        ROUND((POWER(orbit_time, 2) / (POWER(semi_major_axis, 3))),3) AS Kepler_3rd_law_dependency
17 FROM Stars_and_planets2 AS SP2
18 WHERE star_count > 1
19 ORDER BY star_count DESC
20
21 /*kepler's 3rd law states that the semi major axis of the planets orbit and the time of the planets orbit are
22 dependent in a way that always (T^2/R^3) is constant for all the planets that surround a certain star
23 here is the table to check it*/
24

```

91 % Nie znaleziono żadnych problemów W.: 5, Zn.: 43, Kol.: 46 TABULATORY CRLF Windows 1250

Wyniki Komunikaty

star_name	star_count	planet_name	orbit_time	semi_major_axis	T^2	R^3	Kepler_3rd_law_dependency
1 Sun	8	Earth	1	1	1	1	1
2 Sun	8	Jupiter	11.86	5.204	140.6596	140.932729664	0.998
3 Sun	8	Mars	1.88	1.523	3.5344	3.532642967	1
4 Sun	8	Mercury	0.24	0.387	0.0576	0.057960603	0.994
5 Sun	8	Neptune	164.79	30.07	27155.7441	27189.441343	0.999
6 Sun	8	Saturn	29.45	9.582	867.3025	879.768685368	0.986
7 Sun	8	Uranus	84.02	19.218	7059.3604	7097.813228232	0.995
8 Sun	8	Venus	0.61	0.723	0.3721	0.377933067	0.985
9 Alpha Centauri A	3	Proxima Centauri b	0.03	0.048	0.0009	0.000110592	8.138
10 Alpha Centauri A	3	Proxima Centauri c	5.21	1.48	27.1441	3.241792	8.373
11 Alpha Centauri A	3	Proxima Centauri d	0.014	0.029	0.000196	2.4389E-05	8.036

Zapytanie zostało wykonane pomyślnie. sql198169server.database.wi... s198169 (51) s198169-space 00:00:00 Wiersz: 1, kolumna: 1 Wiersze: 11

Query9

```

1  SELECT
2  BOSS.observer_name AS Boss_name,
3  BOSS.observer_surname AS Boss_surname,
4  'is the boss of' AS 'Text',
5  EMPLOYEE.observer_name AS Employee_name,
6  EMPLOYEE.observer_surname AS Employee_surname,
7  'in observatory' AS 'Text', Observatory_name
8  FROM Observer AS EMPLOYEE
9  INNER JOIN Work
10 ON EMPLOYEE.observer_id = Work.observer_id
11 INNER JOIN Observatory
12 ON Work.observatory_id = Observatory.observatory_id
13 INNER JOIN Observer AS BOSS --self join here, however this self join is from a relationship Observer -> is boss in -> Observatory
14 ON Observatory.boss_id = BOSS.observer_id
15 WHERE
16 Work.date_of_end IS NULL
17 AND BOSS.observer_id != EMPLOYEE.observer_id --excludes someone to be their own boss
18 ORDER BY BOSS.observer_id
19
20 -- Who is the boss of who and in which observatories

```

83 % Nie znaleziono żadnych problemów W.: 17, Zn.: 87, Kol.: 93 TABULATORY CRLF Windows 1250

Wyniki Komunikaty

Boss_name	Boss_surname	Text	Employee_name	Employee_surname	Text	Observatory_name
1 Michael	Satellarmy	is the boss of	Sebastian	Alvarez	in observatory	Chandra X-ray Obs.
2 Michael	Satellarmy	is the boss of	Paulina	Konstelarna	in observatory	Spitzer Space Telescope
3 Michael	Satellarmy	is the boss of	Jakub	Kasztan	in observatory	Kepler Space Telescope
4 Michael	Satellarmy	is the boss of	Timothy	Cheese	in observatory	Kepler Space Telescope
5 Nicolaus	Bossy	is the boss of	Marek	Andromedziarz	in observatory	Gaia
6 Nicolaus	Bossy	is the boss of	Ryszard	Ziemiak	in observatory	Lowell
7 Nicolaus	Bossy	is the boss of	Michael	Satellarmy	in observatory	Voyager 2
8 Nicolaus	Bossy	is the boss of	Michael	Satellarmy	in observatory	Gaia
9 Sebastian	Alvarez	is the boss of	Igor	Owczynski	in observatory	Mauna Kea
10 Sebastian	Alvarez	is the boss of	Agata	Gwiezdna	in observatory	Mauna Kea
11 Sebastian	Alvarez	is the boss of	Marek	Mleczodrogi	in observatory	Paranal
12 John	Pork	is the boss of	John	Czarnodziury	in observatory	Roque de los Muchachos
13 John	Pork	is the boss of	Natalia	Planetarna	in observatory	Teide
14 John	Pork	is the boss of	Sebastian	Alvarez	in observatory	Roque de los Muchachos

Zapytanie zostało wykonane pomyślnie. sql198169server.database.wi... s198169 (52) s198169-space 00:00:00 Wiersz: 1, kolumna: 1 Wiersze: 14